Exam 1 :—

(1)

(a) $$\int \frac{6-3x+8x^5}{2x^2}\,dx$$

$$= \int \left[\frac{6}{2x^2} - \frac{3x}{2x^2} + \frac{8x^5}{2x^2}\right] dx$$

$$= \int \left[\frac{3}{x^2} - \frac{3}{2x} + 4x^3\right] dx$$

$$= \left[-\frac{3}{x} - \frac{3}{2}\ln|x| + \frac{4x^4}{4}\right] + C$$

$$= \left[-\frac{3}{x} - \frac{3}{2}\ln|x| + x^4\right] + C$$

$$\therefore \int \frac{6-3x+8x^5}{2x^2}\,dx = \left[-\frac{3}{x} - \frac{3}{2}\ln|x| + x^4 + C\right]$$

(b) $$\int \frac{x^2 \ln(x^3+2)}{(x^3+2)}\,dx$$

$\int \frac{1}{u} = \ln|u|$

$\int \ln(u) =$

Let $u = x^3+2$, $du = 3x^2\,dx$

$$\frac{du}{3} = x^2\,dx$$

$$= \int \frac{\ln(u)}{u}\,du$$

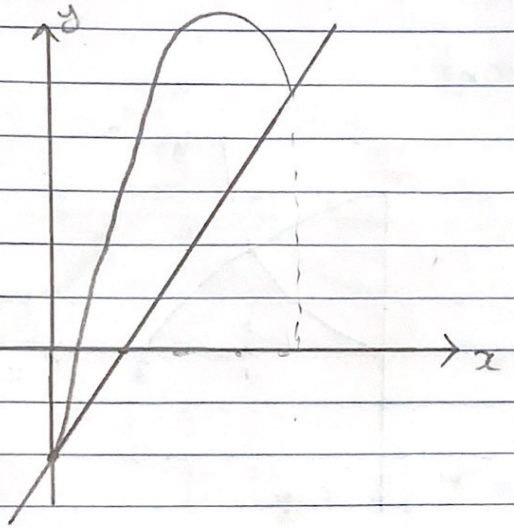
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(c) $\int \frac{x^5}{\sqrt[4]{1+x^3}}\,dx$

Let $u = 1+x^3$ $\quad du = 3x^2\,dx$

$(u-1)^2 = x^5$ $\quad \frac{du}{3} = x^2\,dx$

(2) $y = x - 2$ and $y = 5x - x^2 - 2$



$x - 2 = 5x - x^2 - 2$

$x^2 + x - 5x - 2 + 2 = 0$

$x^2 - 4x = 0$

$x(x-4) = 0$ ✓ $x = 0$, ✓ $x = 4$

$$A(x) = \pi \int_0^4 (5x - x^2 - 2) - (x - 2)\, dx$$

$$= \pi \int_0^4 (5x - x - x^2 - 2 + 2)\, dx$$

$$= \pi \int_0^4 (4x - x^2)\, dx$$

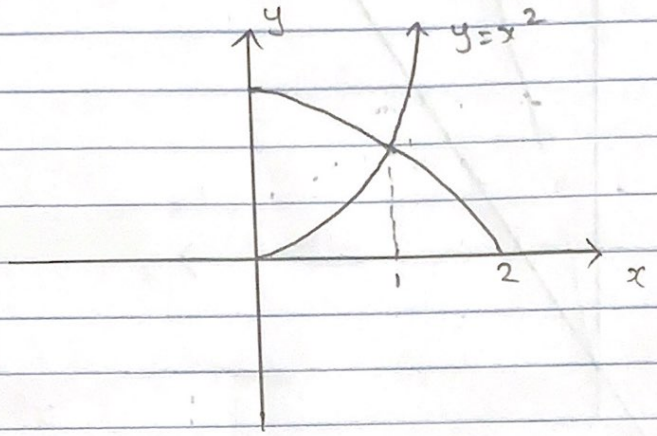
$$= \pi \left[\frac{4x^2}{2} - \frac{x^3}{3} \right]_0^4$$

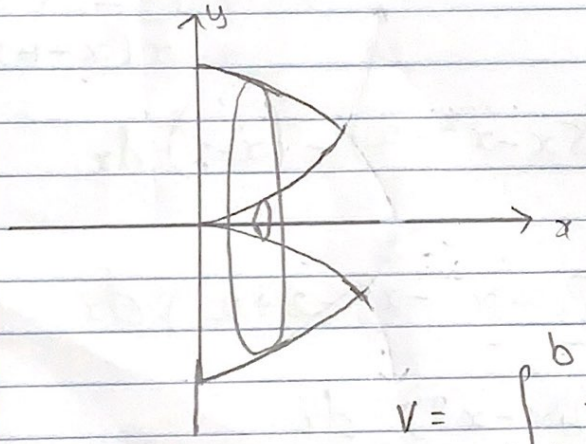
$$= \pi \left[32 - \frac{64}{3} \right] = \pi \left[\frac{32}{3} \right]$$

$$\boxed{A(x) = \frac{32}{3}\pi}$$

(3) $y = x^2$ and $y = \sqrt{2-x}$ is about x-axis

First quadrant





$$V = \int_a^b \pi (\text{outer}^2 - \text{inner}^2)\, dx$$

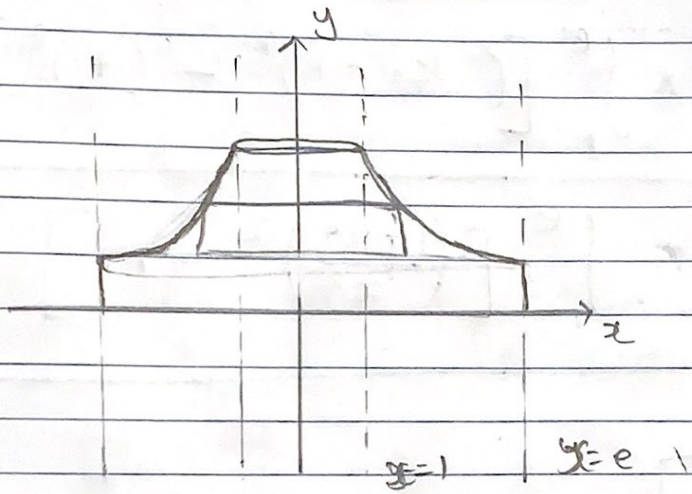
$$V = \int_0^1 \pi \left((\sqrt{2-x})^2 - (x^2)^2 \right) dx$$

$$V = \pi \int_0^1 \left[(2-x) - x^4 \right] dx$$

$$\boxed{V = \pi \int_0^1 (2 - x - x^4)\, dx}$$

(11) $y = \frac{1}{x^2}$, $x = 1$, $x = e$ and $y = 0$

(a)



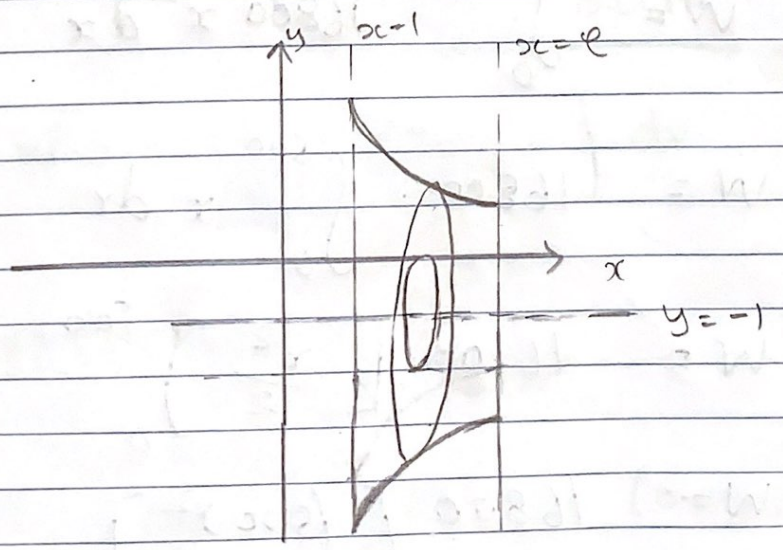
Shell method:-

$$V = \int_a^b 2\pi \,(\text{Radius})\,(\text{Top} - \text{bottom})\,dx$$

$$V = \int_0^e 2\pi (x+1)\left(\frac{1}{x^2}\right) dx$$

$$\boxed{V = 2\pi \int_0^e \left(\frac{x+1}{x^2}\right) dx}$$

(b)

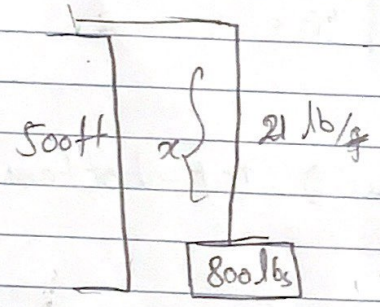


$$V = \pi \int_a^b (\text{outer}^2 - \text{inner}^2)\, dx$$

$$V = \pi \int_1^e \left[\left(\tfrac{1}{x^2} + 1 \right)^2 - (1)^2 \right] dx$$

$$\boxed{V = \pi \int_1^e \left[\left(\frac{1+x^2}{x^2} \right)^2 - 1 \right] dx}$$

(5)



$$W = F * D$$

$$W = \int_0^{500} (800)(21x)\, dx$$

$$W = \int_0^{500} 16800\, x\, dx$$

$$W = 16800 \int_0^{500} x\, dx$$

$$W = 16800 \left[\frac{x^2}{2} \right]_0^{500}$$

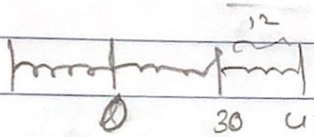
$$W = 16800 \left[\frac{(500)^2}{2} \right]$$

$W = 16800 \left[125000 \right]$

$$W = 2100000000 \text{ J}$$

(6) $2\text{J} \Rightarrow$ 30cm to 42cm

$? \Rightarrow$ 35cm to 40cm



$F = kx \Rightarrow 2 = kx$

$$2 = \int_0^{0.12} kx\,dx \Rightarrow 2 = \left[\frac{kx^2}{2} \right]_0^{0.12}$$

$$H = k\left[(0.12)^2 - 0\right] \Rightarrow H = k(0.0144)$$

$$\boxed{\frac{H}{0.0144} = k} \qquad \boxed{F = \frac{H}{0.0144}(x)}$$

$$W = \int_{0.05}^{0.10} \left[\frac{H}{0.0144} x \right] dx$$

$$W = \left[\frac{2}{0.0144} x^2 \right]_{0.05}^{0.10}$$

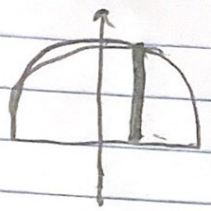
$$W = \frac{2}{0.0144}\left[(0.10)^2 - (0.05)^2\right]$$

$$W = \frac{2}{0.0144}\left[0.01 - 0.0025\right]$$

$$\boxed{W = 1.041666\overline{7}\ J}$$

(7)

$x^2 + y^2 = 100$



$y = \sqrt{100 - x^2}$

Area of square $= y^2$

$$\boxed{V = \int_0^{10} \left(\sqrt{100 - x^2}\right)^2 (1000)(9.8)\, dx}$$

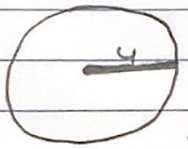
(8)

$$W = \int_a^b \rho \cdot g \,(\text{Area of crossection})\,(\text{distance out of tank})\,dx$$

$$W = \int_0^{10} (1493)(9.8)(A)(D)\,dx$$

$x^2 + y^2 = 16$

$y = \sqrt{16 - x^2}$



$$A = \pi r^2 h \;\rightarrow\; \pi (r)^2 (x)$$

$$W = \int_0^{10} (1493)(9.8)(\pi)\left(\sqrt{16-x^2}\right)^2 x\,dx$$

$$\boxed{W = \int_0^{10} (1493)(9.8)\pi x\,(16 - x^2)\,dx}$$